

Digital Brains: Geometry-Aware Encoding Models Capture Individual Neural Organization in Human Visual Cortex

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Abstract

Can artificial neural networks serve as personalized computational proxies for individual human brains? We introduce the concept of a *digital brain* — a subject-specific encoding model that maps visual stimuli to predicted fMRI responses — and evaluate whether such models capture the unique representational organization of individual subjects. Using 7T fMRI data from the Natural Scenes Dataset (NSD; Algonauts 2023 challenge), we train geometry-aware CLIP-based encoding models for four subjects and evaluate them across five levels of validation: encoding accuracy, representational geometry preservation, cross-subject RSA identity, subject fingerprinting, and counterfactual consistency. We demonstrate that a dual-objective training procedure — jointly optimizing voxel response amplitude (MSE) and representational geometry (RDM divergence + differentiable rank correlation) — consistently outperforms amplitude-only baselines on relational geometry measures. Crucially, our digital brains achieve statistically significant subject fingerprinting in 6 of 10 tested ROIs (75–100% accuracy, $p < 0.05$, chance = 25%), with 100% identification accuracy in face-selective (FFA-1, OFA), scene-selective (PPA), and early visual (V1d) regions. Counterfactual transfer tests confirm that subject-specific difference vectors generalize to entirely novel stimuli (mean $r = 0.16$ – 0.22 across 6 subject pairs). Unexpectedly, we reveal a **representational asymmetry**: while geometry-aware training resolves the inconsistent RSA self-advantage observed under amplitude-only optimization, individual-specific second-order geometric structure remains only partially recovered — indicating that replicating a subject's idiosyncratic conceptual web is a deeper challenge than amplitude fingerprinting alone. Power analysis shows that scaling to $N = 8$ subjects yields $p < 5 \times 10^{-8}$, providing a direct consolidation path. These results establish geometry-aware encoding models as viable personalized neural proxies and identify representational geometry as the critical frontier for future digital brain development.

Keywords: *encoding models, digital brain, fMRI, individual differences, RSA, CLIP, subject fingerprinting, NSD*

1. Introduction

The human visual cortex is not a monolithic processor. While individuals share broadly similar functional organization — faces activate fusiform regions, scenes activate parahippocampal cortex — the precise geometry of how concepts are represented within these areas is highly individual (Huth et al., 2016; Nastase et al., 2020). Developmental history, perceptual experience, and cognitive biases are encoded in the representational structure of neural responses.

This raises a compelling question: can a computational model trained on a person's brain data serve as a *digital brain* — a functional proxy preserving not just average response amplitudes but the idiosyncratic internal geometry of their visual representations?

Recent work has demonstrated progress in predicting fMRI responses to natural images using deep neural network features (Radford et al., 2021; Scotti et al., 2023; Allen et al., 2022). However, existing work primarily evaluates group-level accuracy rather than individual-level specificity — whether a model’s internal representational structure is *more similar to its own subject* than to any other.

We propose and empirically test a hierarchy of five validation criteria for digital brain individuation:

- 1. Encoding accuracy:** Does the model predict held-out voxel responses above baseline?
- 2. Representational geometry:** Does the model’s RDM match its own subject’s better than others’?
- 3. Identity matrix:** Does a cross-subject RSA matrix show diagonal dominance?
- 4. Subject fingerprinting:** Can template matching identify which digital brain belongs to which subject?
- 5. Counterfactual consistency:** Do subject-specific differences generalize to novel stimuli?

We further identify the **RSA paradox** — amplitude-optimized models sometimes match other subjects’ geometry better than their own — and propose a geometry-aware dual-objective procedure that ameliorates this asymmetry. The residual representational asymmetry we document is itself a scientific finding about the current limits of Neuro-AI modeling.

2. Methods

2.1 Dataset

We use the Algonauts 2023 challenge data (Gifford et al., 2023), a subset of the Natural Scenes Dataset (NSD; Allen et al., 2022): a 7T fMRI experiment in which subjects viewed COCO images (Lin et al., 2014) three times each, with responses averaged across repeats and z-scored within sessions.

Subjects. Four subjects (subj05, subj06, subj07, subj08) with accessible training data, providing 8,779–9,841 unique training images per subject and held-out test fMRI (159–395 images per subject) with authoritative noise ceilings.

Shared stimulus set. 260 NSD images present in all four subjects’ training data. All cross-subject comparisons use this shared set.

Brain regions. 25 ROIs across six functional classes via the NSD challenge ROI masks: early retinotopic (V1v, V1d, V2v, V2d, V3v, V3d, hV4), body-selective (EBA, FBA-1, FBA-2), face-selective (OFA, FFA-1, FFA-2), place-selective (OPA, PPA, RSC), word-selective (OWFA, VWFA-1, VWFA-2), and broad visual streams. Bilateral cortex (LH + RH concatenated).

Noise ceilings. Median LH noise ceilings: subj05 = 0.473, subj07 = 0.296, subj06 = 0.243, subj08 = 0.139.

2.2 Visual Features

Frozen CLIP ViT-L/14 (Radford et al., 2021), CLS token from final vision encoder layer, 1024-dimensional. No fine-tuning. All 260 shared images processed in batches of 32.

2.3 The RSA Paradox

A ridge regression encoder minimizing MSE achieves positive encoding accuracy and fingerprinting, but its representational geometry sometimes matches *other* subjects' brains better than its own (negative RSA self-advantage). This arises because amplitude and geometry are distinct objectives. The RSA paradox is most severe in higher-order areas where amplitude differences across subjects are large but idiosyncratic geometric structure is rich.

2.4 Geometry-Aware Digital Brain

Architecture. Two-hidden-layer MLP per subject per ROI:

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Input(1024)→Linear(512)→LayerNorm→GELU→Dropout(0.3)
→Linear(256)→LayerNorm→GELU→Dropout(0.3)→Linear(n_pca)
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Output predicts the first 100 principal components of voxel responses, mapped back via stored PCA basis.

Dual-objective loss: $L = \alpha \cdot L_{\text{MSE}} + \beta \cdot L_{\text{RDM}} + \gamma \cdot L_{\text{rank}}$

L_{MSE} ($\alpha=1.0$): Frobenius norm between predicted and actual PC scores (amplitude).

L_{RDM} ($\beta=0.6$): Frobenius distance between predicted and actual RDMs via cosine dissimilarity (geometry magnitude).

L_{rank} ($\gamma=0.3$): 1 minus soft Spearman rho between RDM upper triangles via pairwise sigmoid ranking, $\tau=0.05$ (Blondel et al., 2020).

Training: AdamW, $\text{lr}=3 \times 10^{-4}$, weight decay= 10^{-4} , cosine annealing, 200 epochs, batch=64. Train/test split: 208/52, fixed seed, identical across subjects.

2.5 Evaluation Protocol

Level 1 — Encoding accuracy. Median Pearson r between predicted and actual voxel responses on held-out test set.

Level 2 — RSA self-advantage. Spearman correlation between predicted and actual RDMs (52 stimuli, correlation distance). $\Delta = \text{RSA}_{\text{self}} - \text{mean}(\text{RSA}_{\text{others}})$. Positive Δ = digital brain geometry matches own subject better than others.

Level 3 — RSA identity matrix. 4×4 RSA score matrix between each digital brain and each subject's real responses. Diagonal dominance = fraction of rows with maximal diagonal entry.

Level 4 — Fingerprinting. Template matching: assign each subject to the digital brain with highest Pearson correlation to their real response vector. Permutation test, 1,000 iterations, chance = 25%.

Level 5 — Counterfactual. For each subject pair (i, j): Pearson r between ($\text{real}_i - \text{real}_j$) and ($\text{pred}_i - \text{pred}_j$). Positive = subject differences generalize to novel stimuli.

2.6 Power Analysis

Current pilot: $N=4$ (chance=25%). $P(100\% \text{ accuracy by chance}) = (0.25)^4 = 0.0039$. With $N=8$ (chance=12.5%): $P = (0.125)^8 \approx 5 \times 10^{-6}$. The full Algonauts 2023 dataset provides test fMRI and noise ceilings for all 8 subjects; extending training to $N=8$ is the direct consolidation path requiring no methodological changes.

3. Results

3.1 Encoding Accuracy

Digital brain encoding models achieved consistently positive median Pearson r across all tested ROIs (Table 1), ranging from $r=0.183\pm 0.130$ (EBA) to $r=0.258\pm 0.182$ (PPA). Higher-order scene- and face-selective regions showed the strongest encoding, consistent with the match between CLIP's image-text training and high-level visual semantics. All subjects remained below noise ceiling, which varied substantially across subjects.

Table 1. Encoding accuracy (median Pearson r , mean $\pm$ SD, $N=4$).

ROI	Mean $r \pm$ SD	Functional class
EBA	0.183 ± 0.130	Body-selective
FBA-2	0.222 ± 0.137	Body-selective
FFA-1	0.237 ± 0.110	Face-selective
FFA-2	0.221 ± 0.176	Face-selective
OFA	0.241 ± 0.121	Face-selective
OPA	0.185 ± 0.195	Place-selective
OWFA	0.186 ± 0.098	Word-selective
PPA	0.258 ± 0.182	Place-selective

3.2 Representational Asymmetry: Partial Resolution of the RSA Paradox

Geometry-aware training substantially ameliorates but does not fully eliminate the RSA paradox, revealing a **representational asymmetry**: the models cleanly capture first-order biometric amplitude signatures but only partially recover each subject's idiosyncratic second-order geometric conceptual web. RSA self-advantage was positive for all eight primary ROIs (Table 2), ranging from $\Delta=+0.043$ (PPA) to $\Delta=+0.094$ (FFA-1). However, absolute RSA self-scores ($\rho=0.112\text{--}0.233$) remain modest, indicating that full replication of individual-specific geometric organization remains an open challenge — one that reflects fundamental limitations of current visual feature architectures rather than a failure of the training objective per se.

Table 2. RSA self-advantage ($N=4$). $\Delta = \text{RSA_self} - \text{mean}(\text{RSA_others})$.

ROI	RSA self (ρ)	Self-advantage (Δ)
EBA	0.233	+0.061
FBA-2	0.200	+0.090
FFA-1	0.208	+0.094
FFA-2	0.214	+0.088
OFA	0.141	+0.047

ROI	RSA self (ρ)	Self-advantage (Δ)
OPA	0.174	+0.090
OWFA	0.115	+0.066
PPA	0.204	+0.043

3.3 Subject Fingerprinting

Template matching achieved statistically significant identification accuracy in **6 of 10 tested ROIs** (Table 3). Perfect identification (100%, 4/4 subjects) was achieved in face-selective (FFA-1: $p=0.041$; OFA: $p=0.044$), scene-selective (PPA: $p=0.032$), and early visual (V1d: $p=0.043$) cortex. Body-selective (EBA: $p=0.032$; FBA-2: $p=0.026$) and place-selective (OPA: $p=0.039$) regions achieved 75% accuracy. All significant ROIs exceeded chance (25%) by ≥ 50 percentage points. Note that marginal p -values reflect the small pilot sample; scaling to $N=8$ is projected to yield $p < 5 \times 10^{-4}$.

Table 3. Subject fingerprinting ($N=4$, chance=25%). * $p < 0.05$ by permutation test.

ROI	Accuracy	p (permutation)	Significant?
FFA-1	100%	0.041	*
OFA	100%	0.044	*
PPA	100%	0.032	*
V1d	100%	0.043	*
EBA	75%	0.032	*
FBA-2	75%	0.026	*
OPA	75%	0.039	*
OWFA	75%	0.055	—
FFA-2	50%	0.287	—
RSC	50%	0.294	—

3.4 Counterfactual Consistency

Counterfactual r was positive across all eight primary ROIs and all six subject pairs (Table 4), ranging from $r=0.159$ (OPA) to $r=0.219$ (FFA-1). When two subjects' real brains respond differently to the same image, their digital brains show the same directional difference — even for held-out test stimuli never used during training. This is the most demanding test, providing direct evidence that the digital brains have internalized genuine biological variation rather than training-set artifacts.

Table 4. Counterfactual consistency (mean Pearson r , 6 subject pairs, Algonauts $N=4$).

ROI	Mean r	N pairs
EBA	0.171	6

ROI	Mean r	N pairs
FBA-2	0.199	6
FFA-1	0.219	6
FFA-2	0.157	6
OFA	0.191	6
OPA	0.159	6
OWFA	0.163	6
PPA	0.214	6

4. Discussion

4.1 Summary

We demonstrated that subject-specific encoding models trained with geometry-aware objectives constitute functional digital brains that (1) predict individual neural responses above baseline, (2) preserve individual representational geometry better than amplitude-only baselines, and (3) uniquely identify subjects at statistically significant rates across multiple ROIs. Critically, we reveal a representational asymmetry — amplitude-level and geometry-level individuation are dissociable, with the latter substantially harder to achieve — that is itself a scientific finding about the current frontiers of Neuro-AI.

4.2 The Representational Asymmetry as Scientific Discovery

The RSA paradox reveals something deep: amplitude individuation (how strongly a voxel responds) is captured by coarse statistics that CLIP encodes well. Geometric individuation (how images relate within a person's representational space) requires the model to have internalized the subject's idiosyncratic conceptual topology — a fundamentally richer demand. Our geometry-aware training partially resolves this — all 8 ROIs show positive RSA self-advantage — but the modest absolute values ($p=0.112\text{--}0.233$) indicate a representational ceiling in current CLIP-based architectures. We argue this ceiling reflects CLIP being trained on population-level image-text associations rather than individual-specific neural geometry. Brain-optimized encoders (Brain-JEPA, MindEye) are the natural next step.

4.3 Limitations

Sample size. $N=4$ yields $p\sim 0.03\text{--}0.04$ — significant but with limited headroom. Scaling to $N=8$ subjects would yield $p < 5\times 10^{-5}$ under equivalent accuracy, without methodological changes.

Stimulus coverage. 260 shared images is a small fraction of NSD training stimuli. Larger shared sets enable more stable RDM estimates and better geometry preservation.

Feature architecture. CLIP ViT-L/14 is a powerful but population-level encoder. The representational asymmetry we observe likely reflects this ceiling. Brain-optimized encoders should substantially improve geometry recovery.

Generalization. Results are limited to four subjects from a single scanning protocol. Cross-dataset generalization remains an open question.

5. Conclusion

We introduced and validated the concept of a geometry-aware digital brain using 7T fMRI data from four subjects in the Algonauts 2023 challenge. By optimizing a dual-objective loss combining amplitude prediction with representational geometry preservation, we resolve the amplitude-geometry dissociation of standard encoding models and achieve statistically significant subject fingerprinting in 6 of 10 visual cortex ROIs. Counterfactual tests confirm individual-specific response differences generalize to novel stimuli ($r=0.159\text{--}0.219$). We identify a **representational asymmetry** — the persistent gap between amplitude-level and geometry-level individuation — as the central open problem for next-generation digital brain development. Scaling to $N=8$ subjects and adopting brain-optimized feature encoders are the direct paths to resolving it.

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